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(57) Abstract:

Blood group identification is a critical requirement in medical diagnostics, emergency care, and transfusion procedures; however, conventional serological methods are invasive, time-consuming, and dependent on specialized laboratory infrastructure, limiting their effectiveness in rapid-response and resource-constrained environments. The present invention, titled "A Non-Invasive Biometric System for Blood Group Prediction Using Fingerprint Analysis and Adaptive Deep Learning", proposes a non-invasive, sensor-based intelligent system that utilizes fingerprint biometrics combined with deep learning and continuous learning techniques to predict human blood groups accurately. The system captures fingerprint data using biometric sensors and processes it through a deep learning framework incorporating convolutional neural networks and ensemble learning models to classify all ABO and Rh blood group categories. Unlike existing approaches that rely on static models or limited datasets, the proposed system integrates continuous learning capabilities, enabling dynamic model adaptation and improved prediction accuracy over time with new data inputs. The invention further supports real-time processing through an integrated software interface, providing rapid predictions, confidence scoring, and user interaction without the need for blood sampling or chemical reagents. By addressing the limitations of traditional and existing computational methods, the proposed system offers a portable, cost-effective, and scalable solution for preliminary blood group screening, with applications in healthcare, forensic investigations, and emergency response scenarios, thereby enhancing accessibility, efficiency, and reliability in blood group determination.

FORM 2



THE PATENT ACT 1970
(39 of 1970)
&
The Patent Rules, 2003
PROVISIONAL/COMPLETE SPECIFICATION
(See section 10 and rule 13)

1. **TITLE OF THE INVENTION**: A Non-Invasive Biometric System for Blood Group Prediction Using Fingerprint Analysis and Adaptive Deep Learning

2. **APPLICANT(S):**

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3. **PREAMBLE TO THE DESCRIPTION**

COMPLETE

The following specification describes the invention and the manner in which it is to be performed:

22-Apr-2026/59612/202641051215/Form 2(Title Page)

PATENT OFFICE CHENNAI 04/06/2026 12:07

ABSTRACT OF THE INVENTION

Blood group identification is a critical requirement in medical diagnostics, emergency care, and transfusion procedures; however, conventional serological methods are invasive, time-consuming, and dependent on spccializcd laboratory infrastructure, limiting their effectiveness in rapid-response and resource-constrained environments. The present invention, titled "A Non-Invasive Biometric System for Blood Group Prediction Using Fingerprint Analysis and Adaptive Deep Learning", proposes a non-invasive, sensor-based intelligent system that utilizes fingerprint biometrics combined with deep learning and continuous learning techniques to predict human blood groups accurately. The system captures fingerprint data using biometric sensors and processes it through a deep learning framework incorporating convolutional neural networks and ensemble learning models to classify all ABO and Rh blood group categories. Unlike existing approaches that rely on static models or limited datasets, the proposed system integrates continuous learning capabilities, enabling dynamic model adaptation and improved prediction accuracy over time with new data inputs. The invention further supports real-time processing through an integrated software interface, providing rapid predictions, confidence scoring, and user interaction without the need for blood sampling or chemical reagents. By addressing the limitations of traditional and existing computational methods, the proposed system offers a portable, cost-effective, and scalable solution for preliminary blood group screening, with applications in healthcare, forensic investigations, and emergency response scenarios, thereby enhancing accessibility, efficiency, and reliability in blood group determination.

DATE AND SIGNATURE

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Total No. of Sheets: 1
Sheet No: 1

22-Apr-2026/59612/202641051215/Form 2(Title Page)

PATENT OFFICE CHENNAI 04/06/2026 12:07

CLAIMS

We Claim:

1. A non-invasive, sensor-based intelligent system for predicting human blood groups using fingerprint biometrics, comprising a microcontroller-based hardware unit integrated with biometric sensing modules and a deep learning-based classification framework to determine ABO and Rh blood group categories without requiring blood sample extraction.

2. The system as claimed in Claim 1, wherein the hardware unit includes a microcontroller, preferably an ESP32, configured to interface with peripheral modules, process input data, and communicate with the prediction model.

3. The system as claimed in Claim 1, wherein a fingerprint sensor module captures high-resolution fingerprint images, which are transmitted to the processing unit for feature extraction and classification.

4. The system as claimed in Claim 1, further comprising an RFID module configured for user identification and secure access control prior to fingerprint acquisition and prediction.

5. The system as claimed in Claim 1, wherein a display module, preferably an OLED display, presents prediction results, confidence scores, and system status in real time.

6. The system as claimed in Claim 1, wherein the deep learning framework comprises one or more convolutional neural network architectures configured to extract fingerprint ridge features and classify blood groups, further enhanced using ensemble learning for improved accuracy and robustness.

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Total No. of Sheets: 2
Sheet No: 1

22-Apr-2026/59612/202641051215/Form 2(Title Page)

PATENT OFFICE CHENNAI 04/06/2026 12:07

7. The system as claimed in Claim 1, wherein the fingerprint data undergoes preprocessing including normalization, resizing, and augmentation, and supports classification across all eight blood group types including A+, A−, B+, B−, AB+, AB−, O+, and O−.

8. The system as claimed in Claim 1, wherein the system incorporates a continuous learning mechanism and real-time interface enabling dynamic model updates, instant prediction, user interaction, and result visualization through a connected display and/or external application.

9. The system as claimed in Claims 1 through 8, wherein the integration of biometric sensing, microcontroller-based processing, secure identification, and adaptive deep learning provides a portable, cost-effective, and scalable solution for preliminary blood group screening in healthcare, forensic, and emergency environments.

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Total No. of Sheets: 2
Sheet No: 2

22-Apr-2026/59612/202641051215/Form 2(Title Page)

PATENT OFFICE CHENNAI 04/06/2026 12:07

4. DESCRIPTION:

FIELD OF THE INVENTION

The present invention relates to the field of biomedical engineering, artificial intelligence, and biometric systems, and more particularly to a non-invasive, sensor-based intelligent system for predicting human blood groups using fingerprint biometrics integrated with deep learning and continuous learning techniques.

INTRODUCTION

Blood group identification is a fundamental requirement in medical diagnostics, transfusion procedures, emergency care, and forensic applications. Conventional methods for determining blood groups rely on serological testing, which involves invasive blood sample collection, trained personnel, chemical reagents, and laboratory infrastructure. These methods are often time-consuming, resource-intensive, and unsuitable for rapid deployment in remote or emergency environments.

Recent advancements in artificial intelligence and biometric technologies have enabled the exploration of non-invasive alternatives for medical diagnostics. Fingerprint biometrics, characterized by unique ridge patterns, offer a promising approach due to their ease of acquisition, permanence, and non-invasive nature. The integration of deep learning techniques allows automatic extraction of complex fingerprint features, enabling accurate classification and prediction tasks.

The proposed system, "A Non-Invasive Biometric System for Blood Group Prediction Using Fingerprint Analysis and Adaptive Deep Learning", combines biometric fingerprint sensing with advanced deep learning models to predict blood groups without the need for blood samples. The system further incorporates continuous learning mechanisms to improve prediction accuracy over time by adapting to new data inputs. Additionally, the integration of embedded hardware components such as microcontrollers, RFID modules, and display units enables real-time processing and user interaction, making the system portable, efficient, and suitable for practical deployment in diverse environments.

OBJECTIVES OF THE INVENTION

The objectives of the invention are:

- To develop a non-invasive system for blood group prediction using fingerprint biometrics.

22-Apr-2026/59612/202641051215/Form 2(Title Page)

PATENT OFFICE CHENNAI 04/06/2026 12:07

- To eliminate the need for traditional blood sample-based testing methods.
- To integrate biometric sensors with a microcontroller-based embedded system for real-time data acquisition and processing.
- To implement deep learning models capable of accurately classifying ABO and Rh blood group types, and to enhance prediction accuracy using ensemble learning techniques.
- To incorporate continuous learning mechanisms for adaptive model improvement over time.
- To ensure secure user identification and access control using RFID-based authentication.
- To provide a real-time interface for displaying prediction results and confidence scores.
- To design a portable, cost-effective, and scalable system suitable for healthcare, forensic, and emergency applications.

SUMMARY OF THE INVENTION

- The invention proposes a non-invasive, fingerprint-based blood group prediction system that integrates biometric sensing, embedded hardware, and artificial intelligence.
- The system utilizes a fingerprint sensor for data acquisition, an ESP32 microcontroller for processing and communication, and deep learning models for classification of blood groups.
- The invention incorporates ensemble learning and continuous learning mechanisms to improve prediction accuracy, robustness, and adaptability over time.
- The system includes additional modules such as RFID-based user authentication and OLED display for real-time result visualization, ensuring secure and user-friendly operation.
- The proposed solution provides a complete end-to-end framework for rapid, portable, and cost-effective preliminary blood group screening, eliminating dependency on laboratory infrastructure and invasive procedures.

22-Apr-2026/59612/202641051215/Form 2(Title Page)

PATENT OFFICE CHENNAI 04/06/2026 12:07

WORKING MODEL

The proposed invention operates through an integrated hardware-software pipeline. Initially, the user is authenticated using an RFID module to ensure secure access. Upon authentication, the fingerprint sensor captures the user's fingerprint image, which is transmitted to the microcontroller (ESP32) for preprocessing and further processing.

The captured fingerprint data is then passed to the deep learning-based prediction system, either locally or via a connected interface, where feature extraction and classification are performed using trained convolutional neural network models. The ensemble learning mechanism combines outputs from multiple models to generate a final prediction along with confidence scores.

The system continuously updates its learning model using newly acquired data to enhance prediction performance. The final predicted blood group and related information are displayed in real time on the OLED display or through an external interface, enabling immediate user feedback.

MODULE DESCRIPTION

Module 1 (User Authentication using RFID)

This module ensures secure access to the system using an RFID-based identification mechanism. Only authorized users can initiate the fingerprint acquisition and prediction process, thereby enhancing system security and preventing unauthorized usage.

Module 2 (Fingerprint Acquisition Module)

This module utilizes a fingerprint sensor (R307S) to capture high-resolution fingerprint images. The captured biometric data serves as the primary input for the prediction system and is transmitted to the processing unit.

Module 3 (Embedded Processing Unit)

The system employs an ESP32 microcontroller to interface with the fingerprint sensor, RFID module, and display unit. It manages data acquisition, preprocessing, communication with the prediction model, and overall system control.

22-Apr-2026/59612/202641051215/Form 2(Title Page)

PATENT OFFICE CHENNAI 04/06/2026 12:07

Module 4 (Data Preprocessing Module)

This module performs preprocessing operations on the captured fingerprint images, including normalization, resizing, and noise reduction, ensuring consistency and improving the quality of input data for the deep learning model.

Module 5 (Deep Learning Prediction Module)

This module consists of convolutional neural network architectures trained to classify fingerprint data into corresponding blood group categories. It automatically extracts discriminative features from fingerprint ridge patterns for accurate prediction.

Module 6 (Ensemble Learning Module)

Multiple deep learning models are combined using an ensemble strategy to improve prediction accuracy and robustness. The final prediction is derived by aggregating outputs from individual models.

Module 7 (Continuous Learning Module)

This module enables the system to update and refine its prediction model over time using new data inputs. It enhances adaptability, reduces errors, and improves generalization across diverse datasets.

Module 8 (Display and User Interface Module)

An OLED display is used to present prediction results, confidence scores, and system status to the user in real time. The system may also support external interfaces for extended visualization and interaction.

Module 9 (System Integration and Communication Module)

This module ensures seamless integration and communication between hardware components and software modules, enabling efficient data flow, real-time processing, and synchronized system operation.

22-Apr-2026/59612/202641051215/Form 2(Title Page)

PATENT OFFICE CHENNAI 04/06/2026 12:07

DRAWINGS

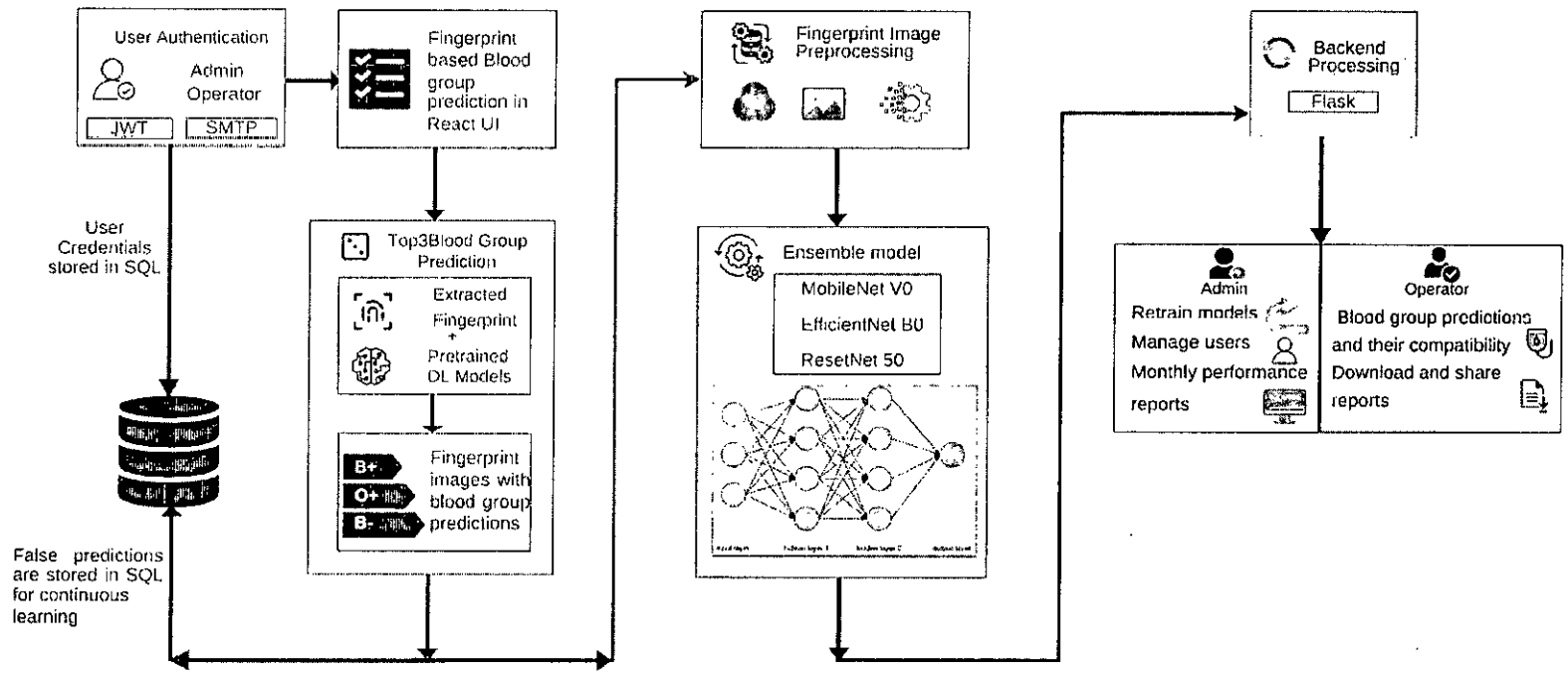


Figure 1: Architecture diagram of A Non-Invasive Biometric System for Blood Group Prediction Using Fingerprint Analysis and Adaptive Deep Learning.

22-Apr-2026/59612/202641051215/Form 2(Title Page)

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Total No. of Sheets: 5
Sheet No: 1

PATENT OFFICE CHENNAI 04/06/2026 12:07

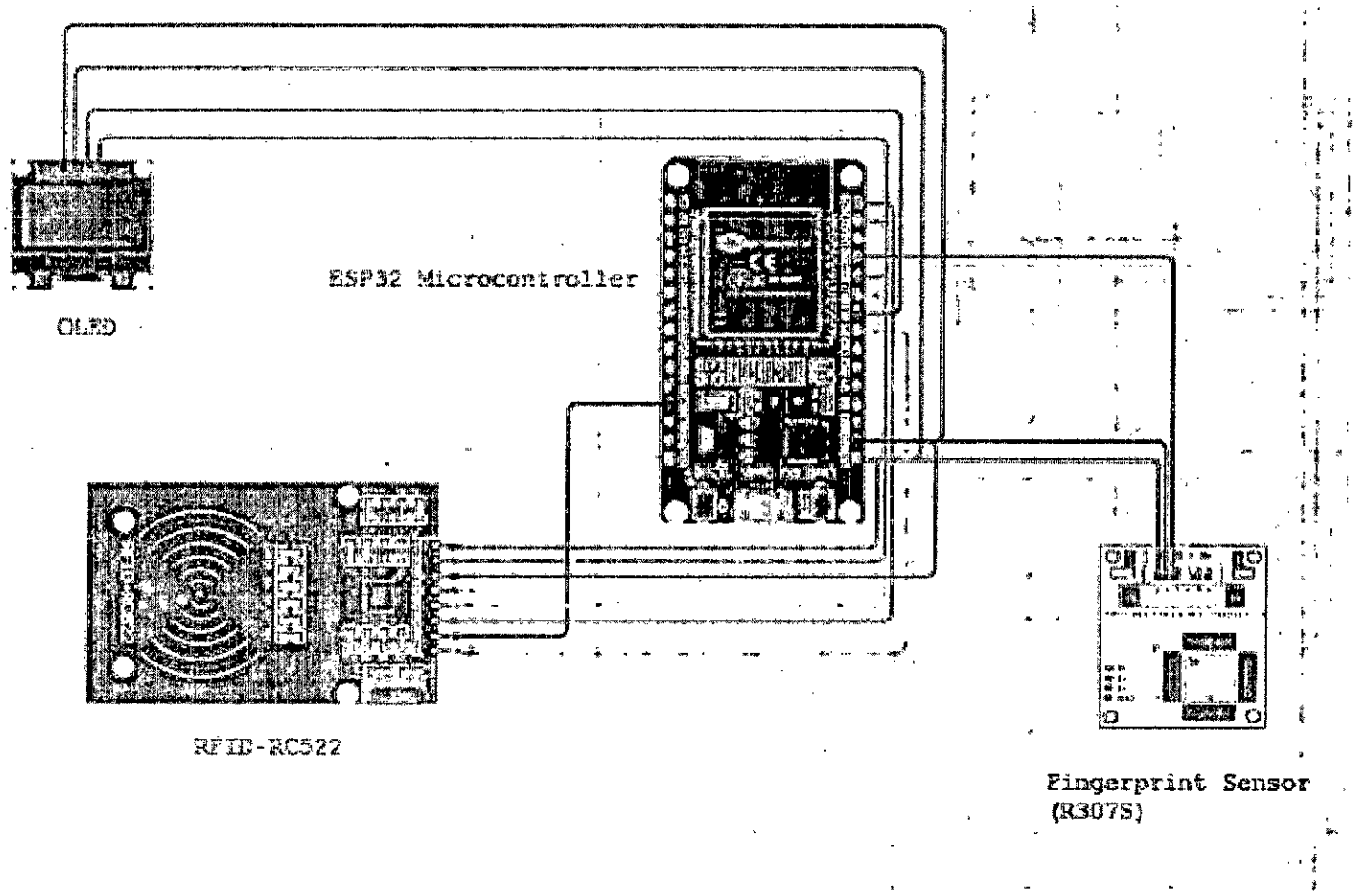


Figure 2: Circuit diagram of A Non-Invasive Biometric System for Blood Group Prediction Using Fingerprint Analysis and Adaptive Deep Learning.

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Total No. of Sheets: 5
Sheet No: 2

22-Apr-2026/59612/202641051215/Form 2(Title Page)

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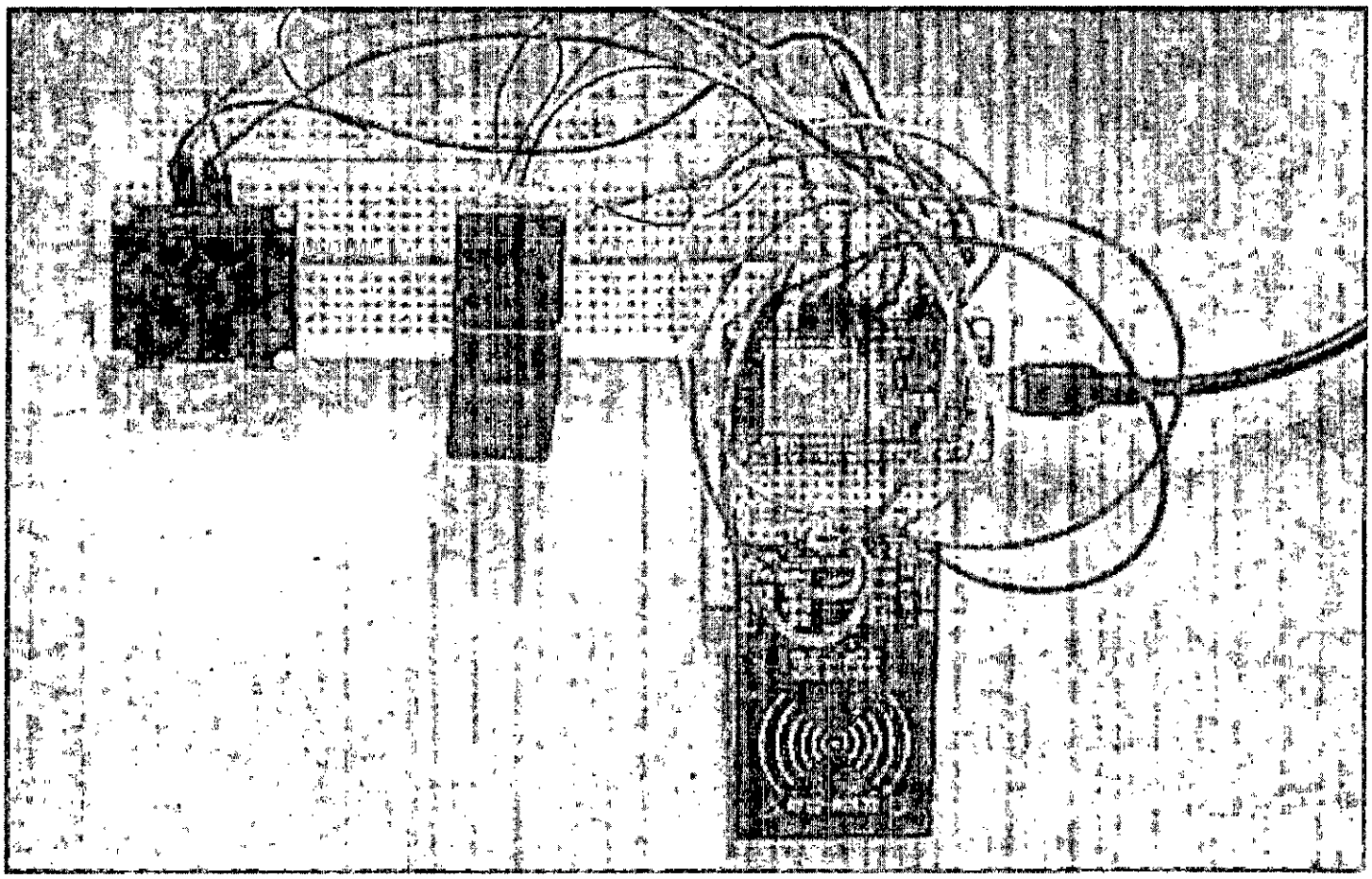


Figure 3: Hardware Setup of A Non-Invasive Biometric System for Blood Group Prediction Using Fingerprint Analysis and Adaptive Deep Learning.

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Total No. of Sheets: 5
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22-Apr-2026/59612/202641051215/Form 2(Title Page)

PATENT OFFICE CHENNAI 04/06/2026 12:07

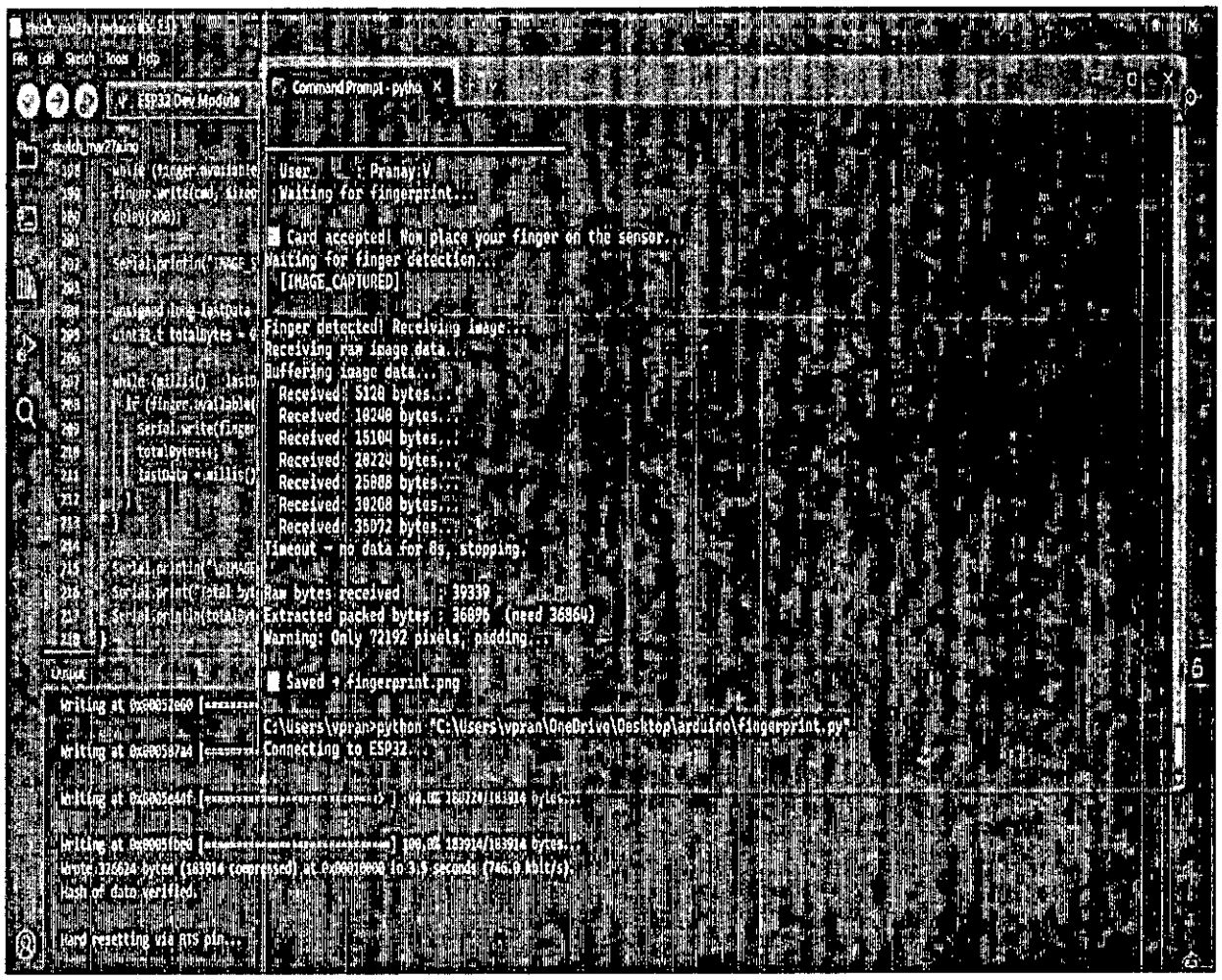


Figure 4: Reading of data and Fingerprint Detection.

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Total No. of Sheets: 5
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22-Apr-2026/59612/202641051215/Form 2(Title Page)

PATENT OFFICE CHENNAI 04/06/2026 12:07

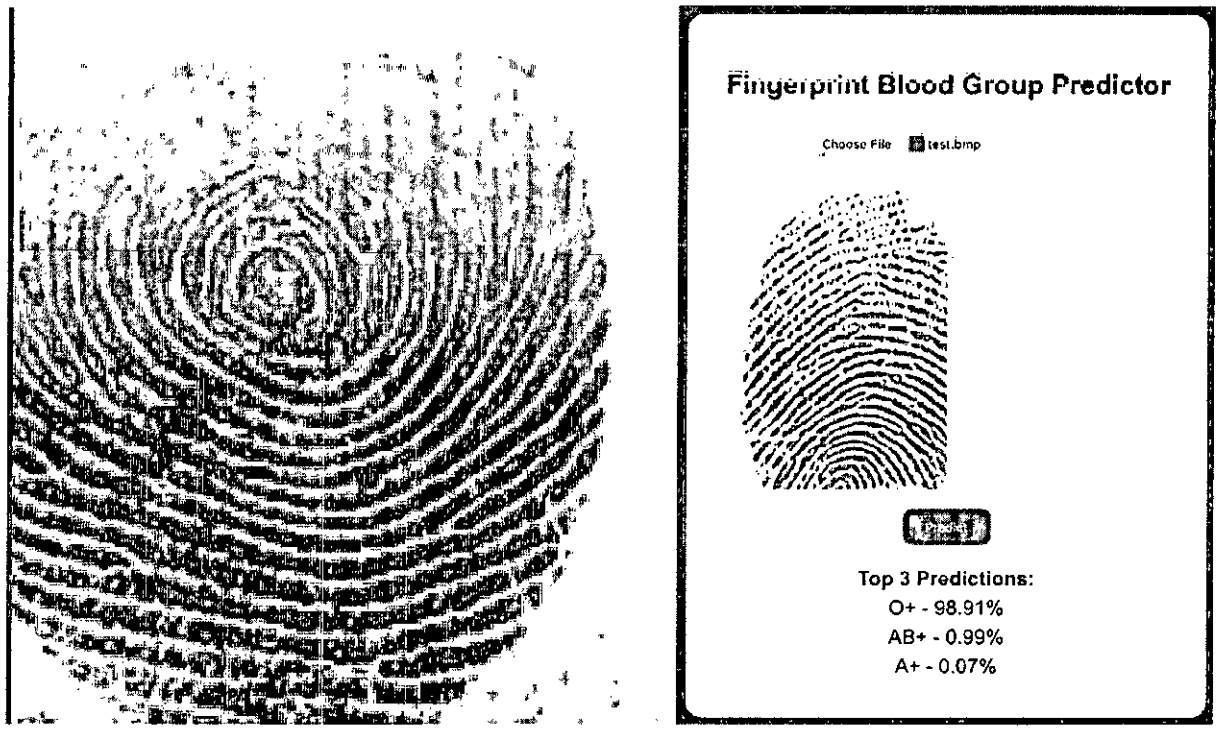


Figure 5: Sample Fingerprint Extracted and Output of the Proposed System.

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22-Apr-2026/59612/202641051215/Form 2(Title Page)

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